Jiawei Xu

Ph.D. Student in Information Studies

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I study the foundations of AI agents, with applications in health.

Publications

Research themes: LLM / agents | Health | Science of Science. Multiple tags indicate overlapping themes.

Conference papers

C1 **SEER: Long-Context Reasoning via Selective Visual-Text Compression**

Xu, J., Zhai, Z., Fang, J., Xu, R., Lu, M., Zhang, Y., Wang, G., Chen, T., & Ding, Y.

2026 / *Conference on Language Modeling (COLM 2026)*. Paper Code

LLM / agents

C2 **MedHallu: A Comprehensive Benchmark for Detecting Medical Hallucinations in Large Language Models**

Pandit, S., **Xu, J.**, Hong, J., Wang, Z., Chen, T., Xu, K., & Ding, Y.

2025 / *EMNLP 2025 Main Conference*. Paper Dataset Code

LLM / agents | Health

C3 **Mapping from Meaning: Addressing the Miscalibration of Prompt-Sensitive Language Models**

Cox, K., **Xu, J.**, Han, Y., Xu, R., Li, T., Hsu, C.-Y., Chen, T., Gerych, W., & Ding, Y.

2025 / *Proceedings of AAAI 2025*. Code Paper

LLM / agents

C4 **From Newborn to Impact: Bias-Aware Citation Prediction**

Lu, M., Wu, M., **Xu, J.**, Li, W., Liu, F., Ding, Y., Sun, Y., Lu, J., & Zhang, Y

2026 / *The Web Conference 2026*. Preprint

LLM / agents | Science of Science

C5 **Interactive Graph Visualization and Teaming Recommendation in an Interdisciplinary Project's Talent Knowledge Graph**

Xu, J., Chen, J., Ye, Y., Sembay, Z., Thaker, S., Payne-Foster, P., Chen, J., & Ding, Y.

2025 / *ASIS&T 2025*. Demo

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Publications (continued)

Conference papers (continued)

C6 **Path-RAG: Knowledge-Guided Key Region Retrieval for Open-ended Pathology Visual Question Answering**

Naeem, A., Li, T., Liao, H.-R., **Xu, J.**, Mathew, A. M., Zhu, Z., Tan, Z., Jaiswal, A. K., Salibian, R. A., Hu, Z., Chen, T., & Ding, Y.

2024 / *ML4H 2024*. Code Paper

LLM / agents | Health

C7 **Decoding Patterns of Data Generation Teams for Clinical and Scientific Success: Insights from the Bridge2AI Talent Knowledge Graph**

Xu, J., Xie, Q., Liu, M., Sembay, Z., Thaker, S., Payne-Foster, P., Chen, J. Y., & Ding, Y.

2024 / *Proceedings of JCDL 2024*.

LLM / agents | Health | Science of Science

C8 **Thought Graph: Generating Thought Process for Biological Reasoning**

Hsu, C.-Y., Cox, K., **Xu, J.**, Tan, Z., Zhai, T., Hu, M., Pratt, D., Chen, T., Hu, Z., & Ding, Y.

2024 / *Companion Proceedings of the ACM Web Conference 2024 (WWW'24 Companion)*. Paper

LLM / agents | Health

Journal articles

J1 **Knowledge Integration and Diffusion Structures of Interdisciplinary Research: A Large-Scale Analysis Based on Propensity Score Matching**

Xu, J., Zheng, Z., Min, C., Huang, W., & Bu, Y.

2025 / *Journal of the Association for Information Science and Technology*. Code Paper

Science of Science

J2 **TAMA: A Human-AI Collaborative Thematic Analysis Framework Using Multi-Agent LLMs for Clinical Interviews**

Xu, H., Yi, S., Lim, T., **Xu, J.**, Well, A., Mery, C., Zhang, A., Zhang, Y., Ji, H., Pingali, K., Leng, Y., & Ding, Y.

2026 / *ACM Transactions on Computing for Healthcare*. Paper

LLM / agents | Health

J3 **PubMed Knowledge Graph 2.0: Connecting Papers, Patents, and Clinical Trials in Biomedical Science**

Xu, J., Yu, C., **Xu, J.**, Torvik, V. I., Kang, J., Sung, M., Song, M., Bu, Y., & Ding, Y.

2025 / *Scientific Data*.

Health | Science of Science

Workshop proposal

2025 **The Second Workshop on GenAI for Health: Potential, Trust, and Policy Compliance**

Xu, J., Xiang, T., Rajpurkar, P., Hong, J., Chen, C., Adeli, E., Li, X., Pavlakos, G., Delp, S., Fei-Fei, L., & Ding, Y

2025 / *NeurIPS 2025 Workshop Proposal*.

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Publications (continued)

Preprints & work under review

P1 **Rethinking the Value of Multi-Agent Workflow: A Strong Single Agent Baseline**

Xu, J., Koesdwiady, A., Bei, S., Han, Y., Huang, B., Wang, D., Chen, Y., Wang, Z., Wang, P., Li, P., & Ding, Y

2026 / *arXiv preprint*. Preprint

LLM / agents

P2 **MATRIX: Multi-Agent Teaming Recommendation through EXpertise Gap Identification**

Xu, J., Ye, Y., Han, Y., Wang, H., Shomer, H., Li, W., Lu, M., Wu, M., Liu, F., Zhang, Y., Sun, Y., Tang, J., & Ding, Y

2026 / *Under review*

LLM / agents | Health | Science of Science

Workshop papers

W1 **Beyond Feature Importance: Auditable Feature Interactions in Predicting Post-Stroke Rigidity with Graph Explainable AI**

Xu, J., Lee, Y., Youssef, A. E., Huang, T., Guo, T., Ying, R., & Ding, Y.

2026 / *DAIH: LLM/VLM Deployment Opportunities and Risks in Healthcare, COLM 2026 Workshop*. Poster. Paper Workshop

Health

W2 **LLM-TA: An LLM-Enhanced Thematic Analysis Pipeline for Transcripts from Parents of Children with Congenital Heart Disease**

Xu, J., Raza, Z., Lim, T., Boddy, L., Mery, C., Well, A., & Ding, Y.

2025 / *GenAI4Health Workshop at AAAI*. Code Paper

LLM / agents | Health

W3 **Position: Open and Closed Large Language Models in Healthcare**

Xu, J., Ding, Y., & Bu, Y.

2024 / *GenAI4Health Workshop at NeurIPS*. Google Scholar

LLM / agents | Health

Education

Aug. 2023–Present
Ph.D. Student in Information Studies
The University of Texas at Austin
iSchool, School of Computing
Committee: Professor Ying Ding (Chair), Professor Yan Zhang, and Professor Kenneth Fleischmann.

Sep. 2021–Jul. 2023
M.S. in Information Science
Peking University
Advisors: Professor Yi Bu and Professor Win-bin Huang.
Excellent Graduate of Beijing

Sep. 2017–Jul. 2021
B.M. in Information Management & Systems
Peking University

Employment

May–Aug. 2026
Applied Scientist Intern
Amazon / Seattle, WA
Self-improving automated research loop for machine learning tasks.
Mentors: Dihan Dai and Keyang He; Manager: Jinhe Shi. Technical advisors: Jiajian Lu and Olcay Boz.

May–Nov. 2025
Applied Scientist Intern
Amazon Search / Palo Alto, CA (SFO24)
Rethinking the Value of Multi-Agent Workflow: A Strong Single Agent Baseline. Featured as the cover paper in DAIR.AI Top AI Papers of the Week.
Mentors: Arief Koesdwiady, Sisong Bei, and Yan Han; Manager: Yutong Chen.

Aug. 2023–Aug. 2026
Graduate Research Assistant
The University of Texas at Austin
iSchool, School of Computing
SELECTED PROJECTS

- **Patient Simulator** Demo
  Developed a voice-based AI patient simulator to help medical students practice clinical conversations and patient communication. Related exploration: SEER, long-context reasoning through selective visual-text compression.
- **Cell Maps for AI (CM4AI)** Demo
  Developed interactive knowledge graph visualization and teaming recommendation tools to help researchers explore expertise and identify potential collaborators.

Aug. 2023–
Aug. 2026

Graduate Research Assistant (continued)

The University of Texas at Austin

iSchool, School of Computing

SELECTED PROJECTS (CONTINUED)

- **LLM uncertainty & hallucination detection for health**

  Core development contributions to Mapping from Meaning, studying prompt sensitivity and uncertainty calibration, and MedHallu, a benchmark for detecting medical hallucinations.
- **LLM-assisted thematic analysis** Paper Code

  Developed methods for identifying themes in interviews with parents of children with congenital heart disease, with evaluation against human analysis and expert review.

Academic service

Workshop leadership

2026 **Workshop Organizer (Student Lead) & Program Co-Chair**
GenAI4Health Workshop at NeurIPS 2026
Sydney, Australia

2025 **Workshop Organizer (Lead) & Program Co-Chair**
GenAI4Health Workshop at NeurIPS 2025
San Diego, CA, USA

2024 **Onsite Student Organizer**
GenAI4Health Workshop at NeurIPS 2024
Vancouver, Canada

Reviewing

Conference PC member & reviewer

ICML; NeurIPS; ACL Rolling Review; ML4H; AAAI.

Journal reviewer

ACM Computing Surveys; Data and Information Management; Journal of the Association for Information Science and Technology; Nature Humanities & Social Sciences Communications; Online Information Review; Data Intelligence; Information & Culture; Journal of Clinical and Translational Science; Scientometrics.

Invited talks

Oct. 2026 — **AI Agents over Biomedical Knowledge Graphs: Finding Collaborators and Querying Data from Research Papers**
NSF Convergence Accelerator meeting
Project demo

Apr. 2025; Oct. 2024 — **CM4AI Talent Knowledge Graph**
Virtual talk

Aug. 2023 — **Beyond Correlation: What Factors Influence Scientific Performance**
Virtual talk

Teaching

2024–2025 — **LLM Prompt Engineering and Multi-Agent Systems**
Guest lecturer, The University of Texas at Austin

2021–2023 — **Data Visualization**
Teaching assistant, Peking University

2022 — **Complex Networks**
Teaching assistant, Peking University

Technical expertise

Languages	English (professional); Mandarin (native).
Everyday workflow	Codex + Claude Code, with me in the loop, taking ideas through research, experiments, and live applications.
Agent harnesses	Harness improvement, skills, prompting, and memory.
LLM post-training	Supervised fine-tuning (SFT) and reinforcement learning (RL) for LLM agents.